

5.2 Zvolíme vhodnou bázi B prostoru $\mathbb{R}^3$.

Podprostor

$$\left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x - y + z = 0 \right\} \quad \text{zapišme jako } \text{span} \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}$$

$$\text{Potom pro } B = \left(\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right)$$

$$\text{je } [f_A]_B^B = \left(\left[\begin{pmatrix} -2 \\ 2 \end{pmatrix} \right]_B, \left[\begin{pmatrix} 2 \\ 1 \end{pmatrix} \right]_B, \left[\begin{pmatrix} 3 \\ 3 \end{pmatrix} \right]_B \right) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$\text{Ale chceme } A = [f_A]_{u_3}^{u_3} = [\text{id}]_{u_3}^B [f_A]_B^B [\text{id}]_{u_3}^{u_3} = \begin{pmatrix} -1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 1 \end{pmatrix}$$

$$\left[A = \begin{pmatrix} 3 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 3 \end{pmatrix} \right]$$

$$\dots \det(A - \lambda I_3) = \det \begin{pmatrix} 3-\lambda & -1 & 1 \\ 1 & 1-\lambda & 1 \\ 1 & -1 & 3-\lambda \end{pmatrix} = (3-\lambda)(2-\lambda)^2$$

$$\text{dále } A \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix} = 3 \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\text{a } M_2 = \ker \begin{pmatrix} 1 & -1 & 1 \\ 1 & -1 & 1 \\ 1 & -1 & 1 \end{pmatrix} = \ker \begin{pmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}$$